

Analyst-Revisions ETF Signal

Strategy

Research Blueprint v2.0

Firm-normalized, novelty-aware, stock-first, point-in-time

English Edition

Version	2.0, regenerated after a formal assessment of the original analyst-consensus plan
Date	August 22, 2026
Primary use	Pre-registration, data engineering, point-in-time backtesting, QuantConnect/LEAN implementation, and ensemble integration
Status	Research specification; no live-trading authority

This document is for research and educational use. It is not investment advice, a performance claim, or an instruction to trade.

Document Purpose

This blueprint replaces the original analyst-rating ETF plan with a more defensible second-generation design. The original research discipline remains: point-in-time data, immutable experiments, pre-registration, walk-forward testing, untouched holdouts, realistic execution, and explicit leverage gates. The signal model is upgraded in six major ways:

1. normalize ratings within each brokerage firm's own ordered scale;
2. measure consensus-relative novelty rather than treating every revision as equally informative;
3. separate rating, price-target, EPS-estimate, initiation, and reiteration channels;
4. model breadth, institution independence, and common-event clustering explicitly;
5. use sector-relative stock scores and reliability-shrunk ETF aggregation;
6. compare direct-stock, conventional industry, and ETF aggregation before claiming that the ETF topology adds value.

Canonical research question

Does point-in-time, stock-first aggregation of firm-normalized and novelty-aware analyst revisions predict future ETF excess returns beyond momentum, earnings information, sector trends, style exposures, and simple industry aggregation?

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Executive Research Specification

1.1 Canonical question

The research program tests whether analyst information can be converted into a tradable ETF-ranking signal without mistaking public-event reactions or price momentum for independent analyst alpha.

Primary hypothesis

ETFs with greater point-in-time exposure to stocks experiencing recent, firm-normalized, consensus-novel, and independently corroborated favorable analyst revisions will subsequently outperform peer ETFs with weaker or adverse analyst evidence, after controlling for momentum, earnings, sector, style, volatility, and liquidity.

Let i index stocks, j ETFs, e analyst events, and t decision times. The V2 research chain is:



1.2 Three forms of value

The strategy may create value through:

1. **Selection alpha:** top-ranked ETFs outperform lower-ranked peers.
2. **Avoidance alpha:** adverse analyst clusters identify exposures that deserve less capital even when absolute returns remain positive.
3. **Portfolio-quality improvement:** ETF aggregation reduces company-specific error relative to direct ownership of seed stocks.

The most realistic prior is not that analyst ratings become a standalone high-alpha strategy. The more defensible expectation is a medium-strength selection and theme-discovery engine that improves an ensemble when its errors are imperfectly correlated with insider-buying and short-interest engines.

1.3 Canonical V2 score at a glance

The preferred confirmatory model is intentionally simpler than the full feature library:

$$\Delta r_e = r_{f(e)}(g_e^{new}) - r_{f(e)}(g_e^{old}), \quad (1.1)$$

$$A_{i,t}^{raw} = \sum_{e \in \mathcal{E}_{i,t}} \Delta r_e 2^{-d_e/20}, \quad (1.2)$$

$$Z_{i,t}^A = Z_{\text{robust,sector}(i),t}(A_{i,t}^{raw}), \quad (1.3)$$

$$\tilde{A}_{i,t} = Z_{i,t}^A \rho_{i,t}, \quad (1.4)$$

$$\bar{S}_{j,t} = \frac{\sum_i w_{ij,t} \tilde{A}_{i,t}}{\max(C_{j,t}, \varepsilon)}, \quad (1.5)$$

$$S_{j,t}^{V2} = \bar{S}_{j,t} \sqrt{C_{j,t}} \min \left(1, \sqrt{\frac{N_{j,t}^{eff}}{5}} \right). \quad (1.6)$$

Consensus novelty, price-target revisions, EPS revisions, breadth, and analyst quality are retained as independent diagnostics or later-stage extensions. They are not silently multiplied into V2.

Assessment of the Existing Plan

2.1 What should be preserved

The original plan has a strong research foundation. It already requires point-in-time data, pre-registration, immutable experiment artifacts, security-master handling, realistic execution, ETF overlap controls, momentum and earnings confounders, walk-forward tests, an untouched holdout, and a separate leverage phase. Those controls remain mandatory.

2.2 What needs to change

The signal construction is less mature than the research protocol. Its primary weaknesses are:

Weakness	Why it matters
Global rating scale	Different firms use different numbers of ordered categories and different semantics.
Raw revision only	A revision that merely catches up with consensus may contain less information than a contrarian revision.
Mixed event types	Initiations, reiterations, target-only actions, and true upgrades are economically different.
Universal half-life	Post-earnings target changes and non-earnings rating changes can decay on different clocks.
Hard ETF coverage threshold	A 50% rule ignores whether one stock dominates the observed signal.
Insufficient common-event control	Fifteen analysts reacting to one earnings release are not fifteen independent pieces of underlying news.
No topology benchmark	The plan does not first prove that ETF aggregation beats direct stock or industry aggregation.
Arbitrary weekly clock	Analyst events arrive continuously; Monday-only rebalancing introduces uneven latency.

2.3 Decision

Assessment

Retain the original research controls. Replace the stock signal and ETF aggregation layers. Treat the revised plan as Version 2.0 rather than as an unrelated strategy.

Hypotheses and System Boundaries

3.1 Pre-registered hypothesis family

The research program separates claims that are often conflated.

3.1.1 H1: Static consensus

$$H_1 : \text{Consensus}_{i,t} \text{ predicts } R_{i,t \rightarrow t+h}.$$

This is a benchmark, not the preferred model.

3.1.2 H2: Rating revisions

$$H_2 : \Delta r_e \text{ predicts future excess returns.}$$

3.1.3 H3: Consensus novelty

$$H_3 : u_e = r_e^{new} - \bar{r}_{i,t}^{(-a)}$$

adds information beyond the analyst's own revision.

3.1.4 H4: Independent breadth

Several institutions moving in the same direction over a short interval contain more information than one institution repeating the same view.

3.1.5 H5: ETF topology

$$H_5 : IC(S^{ETF}, R_{future}^{ETF}) > IC(S^{Industry}, R_{future}^{Industry})$$

or ETF aggregation materially improves Sharpe, drawdown, turnover, or implementation quality relative to direct seed stocks.

3.1.6 H6: Incremental ensemble value

The analyst engine retains predictive or risk-management value after controlling for insider buying, short interest, momentum, earnings information, and market regime.

3.2 V1 exclusions

The confirmatory implementation excludes:

- end-to-end machine learning over raw vendor fields;
- leverage or inverse ETFs based solely on score magnitude;
- intraday trading that assumes privileged brokerage distribution;
- discretionary rating-map changes after observing returns;
- automatic analyst-quality weights without hierarchical shrinkage;
- unsupported historical consensus reconstruction before a warm-up period.

No hedge camouflage

Evaluate stock signals, ETF rankings, and the unhedged analyst portfolio before any hedge overlay. A hedge can improve historical Sharpe by removing market beta without proving that analyst evidence predicts cross-sectional ETF returns.

Data Source Architecture

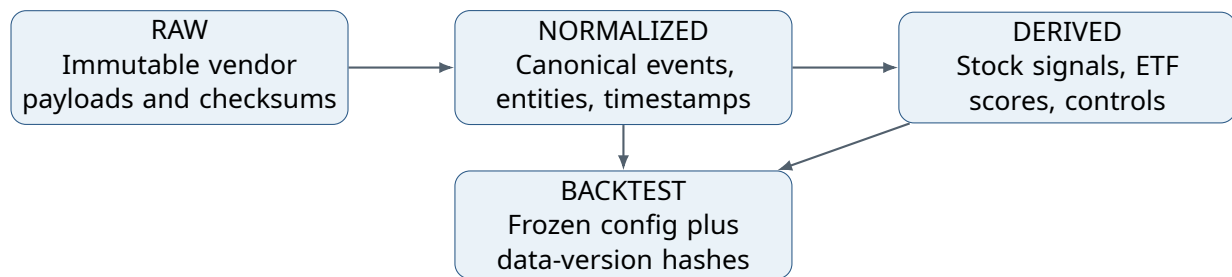
4.1 Preferred division of responsibilities

The implementation should be vendor-agnostic after normalization.

Dataset	Preferred role	Point-in-time requirement
Analyst events	Benzinga or equivalent event-level vendor	Publication timestamp, prior/current rating, firm, analyst, action, prior/current target; immutable snapshot
ETF constituents	QuantConnect US ETF Constituents	Historical constituent weights, EndTime, LastUpdate, conservative publication lag
Security master	QuantConnect US Equity Security Master plus vendor identifiers	Symbol changes, corporate actions, permanent security identity
Prices	QuantConnect equity/ETF prices	Split/dividend handling and realistic next-open execution
Earnings/events	Point-in-time earnings and corporate-event calendar	Actual public availability, not retrospectively corrected dates
Factor controls	Market, sector, momentum, beta, volatility, style	Computed only from information available before the decision time

Benzinga’s current product page states that its analyst-rating history begins in 2013, covers US equities, includes intraday timestamps, current and prior ratings and targets, analyst or firm identity, action type, and unnormalized sell-side terminology [1]. QuantConnect currently documents 2,650 US ETF listings, coverage beginning in June 2009, daily resolution from January 2015 and monthly resolution earlier, with a possible delay of up to one week [2].

4.2 Raw, normalized, and derived layers



The backtester must never call the vendor API, infer rating semantics, resolve current tickers, or revise event timestamps during a run.

Point-in-Time Data Contract

5.1 Two clocks for every observation

Every relevant object must preserve:

$$(effective_time, available_time).$$

The strategy may use observation x at decision time t only if:

$$available_time(x) \leq t.$$

Examples:

- a rating action dated Tuesday at 14:23 cannot be traded at Tuesday's open;
- an ETF holding effective Monday but publicly captured Thursday is unavailable Monday through Wednesday;
- a corrected vendor record cannot replace the original observation unless the correction itself is timestamped and versioned;
- analyst skill at t may use only recommendation outcomes known before t .

5.2 Conservative execution cohorts

Availability quality	Primary execution assumption
Exact premarket timestamp	Next market open
Exact intraday timestamp	Next trading-day open
Exact after-close timestamp	Next trading-day open
Date plus market-session flag	Next open after the stated session
Date only	Next trading-day open plus one conservative day

Unknown or internally inconsistent	Exclude from confirmatory sample
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5.3 Data-validity invariant

An experiment is automatically marked INVALID-DATA if any material PIT test fails. Performance statistics are not reported as evidence when the input clock is invalid.

Security Master and Entity Resolution

Tickers are labels, not permanent identities. The canonical key is a stable `security_id` linked to time-varying vendor and market identifiers.

6.1 Minimum security-master fields

Field	Purpose
<code>security_id</code>	Permanent internal identifier
<code>ticker</code> , <code>valid_from</code> , <code>valid_to</code>	Historical ticker intervals
<code>isin</code> , <code>figi</code> , <code>cik</code> , vendor IDs	Cross-source reconciliation
<code>share_class_id</code>	Prevent incorrect class-level joins
<code>issuer_id</code>	Aggregate issuer-level events only when justified
<code>exchange</code> , <code>country</code> , <code>security_type</code>	Universe eligibility
<code>delisting_date</code> , <code>successor_id</code>	Survivorship and merger handling

6.2 Mapping requirements

The ETF-level candidate set should meet at least:

$$\frac{\sum_{i \in \text{mapped}} w_{ij,t}}{\sum_i w_{ij,t}} \geq 0.99$$

for candidate ETF weight. Unmapped micro-positions may be tolerated if the missing portfolio weight is explicitly reported. Current-ticker joins are prohibited.

Firm-Specific Rating Ontology

7.1 Why one global map is inadequate

Brokerage firms do not share one universal rating vocabulary. Benzinga explicitly states that its ratings retain the language used by each firm rather than being normalized [1]. A three-level and a five-level system should not assign the same economic distance to a one-notch move.

7.2 Ordered firm scale

For firm f , let K_f be the number of valid ordered rating categories during a specified date interval. Let $rank_f(g) \in \{1, \dots, K_f\}$, from most bearish to most bullish. Define:

$$r_f(g) = 2 \frac{rank_f(g) - 1}{K_f - 1} - 1. \quad (7.1)$$

Examples:

Scale	Bearish	Neutral	Bullish
Three levels	-1	0	+1
Five levels	-1, -0.5	0	+0.5, +1

The rating revision is:

$$\Delta r_e = r_{f(e)}(g_e^{new}) - r_{f(e)}(g_e^{old}). \quad (7.2)$$

7.3 Mapping governance

The mapping table must contain:

- firm and valid date range;
- raw label and ordered rank;
- whether the label is company-relative, sector-relative, or absolute;

- mapping confidence and reviewer;
- source evidence and version number.

Ambiguous terms such as *Positive*, *Accumulate*, *Sector Perform*, and *Top Pick* are quarantined until reviewed. The original $\{-2, -1, 0, 1, 2\}$ map remains a baseline for comparison.

Event Taxonomy and Canonical Event Table

8.1 Event types

Every raw record must be classified as:

$$a_e \in \{\text{upgrade, downgrade, initiation, resumption, reiteration, maintain, target-only, coverage-termination}\}. \quad (8.1)$$

Do not convert an initiation into a fictitious neutral-to-buy revision. Do not treat a maintained Buy with a 30% target increase as a zero-information event.

8.2 Canonical event schema

Field family	Required fields
Identity	event ID, security ID, issuer ID, analyst ID, firm ID
Timing	event timestamp, publication timestamp, available timestamp, market session, vendor update timestamp
Action	action type, current/prior raw rating, current/prior firm-normalized rating
Targets	prior/current target, target currency, price immediately before event
Estimates	prior/current EPS estimate, horizon, fiscal period, consensus before event
Context	earnings proximity, guidance, 8-K, price gap, industry-event ID
Quality	timestamp quality, mapping quality, duplicate lineage, correction/amendment lineage
Provenance	source, endpoint, retrieval time, raw-payload hash, data snapshot version

8.3 Deduplication

At most one canonical contribution per institution-security-trading-day enters the V2 rating-revision sum. Raw records are never deleted. They are linked to a normalized economic event through lineage fields.

Historical Consensus Reconstruction

9.1 Active rating rule

For analyst or firm a , stock i , and time t , the last known rating remains active only while:

$$Active_{a,i,t} = \mathbb{I}[0 \leq t - \tau_{a,i} \leq L], \quad (9.1)$$

where $\tau_{a,i}$ is the last valid publication time and L is a maximum age. Test:

$$L \in \{90, 180, 365\} \text{ trading days.}$$

An explicit termination, replacement, or new rating ends the old record sooner.

9.2 Freshness-weighted consensus

$$Consensus_{i,t} = \frac{\sum_a Active_{a,i,t} r_{a,i,\tau_a} 2^{-(t-\tau_a)/H_C}}{\sum_a Active_{a,i,t} 2^{-(t-\tau_a)/H_C}}. \quad (9.2)$$

Static consensus is a benchmark because it is sensitive to stale ratings and left-censored history. Use a 6- to 12-month warm-up before evaluating any reconstructed consensus.

9.3 Coverage state

Store:

- active analyst count;
- active institution count;
- median and maximum rating age;
- share of active ratings with exact timestamps;
- consensus dispersion.

A high average rating with two stale contributors should not be equated with a fresh consensus supported by twenty institutions.

Consensus Novelty and Surprise

Let $a(e)$ be the event analyst or firm. Compute the pre-event consensus excluding that contributor:

$$\bar{r}_{i,t^-}^{(-a)} = \frac{\sum_{\ell \neq a} \eta_{\ell,i,t} r_{\ell,i,t^-}}{\sum_{\ell \neq a} \eta_{\ell,i,t}}, \quad \eta_{\ell,i,t} = 2^{-age_{\ell,i,t}/H_C}. \quad (10.1)$$

Define novelty:

$$u_e = r_{f(e)}(g_e^{new}) - \bar{r}_{i,t^-}^{(-a)}. \quad (10.2)$$

Preserve revision and novelty separately:

Δr_e = change from the same contributor's prior view,

u_e = distance from the market's active analyst consensus.

10.1 Pre-registered novelty models

$$M_1 : \Delta r_e, \quad (10.3)$$

$$M_2 : u_e, \quad (10.4)$$

$$M_3 : \Delta r_e + \lambda_u u_e, \quad (10.5)$$

$$M_4 : \Delta r_e \mathbb{I}[\text{sign}(\Delta r_e) = \text{sign}(u_e)], \quad (10.6)$$

$$M_5 : \Delta r_e \times |u_e|. \quad (10.7)$$

Only M_1 is canonical V2. The others are separate experiments. λ_u may be fixed or learned only inside nested training windows.

Rating, Price-Target, and EPS Subsignals

11.1 Rating revision channel

For true upgrades and downgrades:

$$x_e^R = \Delta r_e. \quad (11.1)$$

11.2 Initiation channel

An initiation has no valid prior neutral state. Define:

$$x_e^{Init} = r_{f(e)}(g_e^{new}) - \bar{r}_{i,t^-}^{(-a)}. \quad (11.2)$$

Analyze initiations separately from upgrades.

11.3 Price-target revision channel

Normalize a target change by the last tradable price before publication:

$$x_e^{PT} = \frac{PT_e^{new} - PT_e^{old}}{P_{i,t^-}}. \quad (11.3)$$

Also retain target-implied upside:

$$Upside_e^{new} = \frac{PT_e^{new} - P_{i,t^-}}{P_{i,t^-}}. \quad (11.4)$$

Winsorize within sector and calendar year using parameters estimated from the training sample only.

11.4 EPS-estimate revision channel

For fiscal horizon h :

$$x_{e,h}^{EPS} = \frac{EPS_{e,h}^{new} - EPS_{e,h}^{old}}{\max(|EPS_{i,h,t^-}^{consensus}|, \varepsilon)}. \quad (11.5)$$

If point-in-time estimates are incomplete, omit this channel rather than reconstructing them from revised history.

11.5 Consistency and disagreement

Define a sign-consistency indicator:

$$Consistent_e = \mathbb{1}[\text{sign}(x_e^R) = \text{sign}(x_e^{PT}) = \text{sign}(x_e^{EPS})]. \quad (11.6)$$

Do not automatically reward consistency. Test whether aligned revisions outperform cases in which a Buy rating is maintained while estimates deteriorate.

Freshness, Decay, and Event Timing

12.1 Exponential decay

For event age d in trading days and half-life H_a :

$$f_a(d) = 2^{-d/H_a} = \exp\left(-\frac{\ln 2}{H_a}d\right). \quad (12.1)$$

Pre-register a small event-specific battery:

Event channel	Candidate half-lives
Rating upgrade/downgrade	5, 20, 60 trading days
Initiation/resumption	20, 60 trading days
Price-target revision	5, 20, 40 trading days
EPS-estimate revision	20, 60, 120 trading days

The canonical rating model keeps $H_R = 20$ trading days. Other values are distinct pre-registered specifications, not tuning knobs applied after reading the holdout.

12.2 Immediate price response

Measure the return from the last pre-event price to the first executable price:

$$Jump_e = \frac{P_{entry,e} - P_{pre,e}}{P_{pre,e}}. \quad (12.2)$$

Run conditional tests for low, medium, and high $|Jump_e|$. A recommendation may contain information but leave little implementable profit after the initial public reaction documented in prior research [3].

Breadth, Independence, and Common-Event Clustering

13.1 Stock-level direction breadth

Over window W :

$$B_{i,t}^A = \frac{N_{i,t}^{upgrade} - N_{i,t}^{downgrade}}{\max(N_{i,t}^{upgrade} + N_{i,t}^{downgrade}, 1)}. \quad (13.1)$$

Record unique analysts and unique institutions separately.

13.2 Effective contributor count

Let $v_e = |x_e^R f_R(d_e)|$. Then:

$$N_{i,t}^{eff} = \frac{(\sum_e v_e)^2}{\sum_e v_e^2}. \quad (13.2)$$

This equals one when one event dominates and increases as independent contributions become more balanced.

13.3 Common-event clustering

Attach a common-event identifier for:

- earnings and guidance;
- 8-K or other material filings;
- investor days and conferences;
- acquisitions and product announcements;
- large overnight gaps;
- industry-wide news.

Define:

$$EventDiversity_{i,t} = \#\{\text{distinct underlying event IDs contributing in } W\}. \quad (13.3)$$

Fifteen analysts reacting to one earnings release may produce high intensity but low event diversity. Retain both features.

Analyst Quality, Skill, and Stickiness

14.1 Why quality must be hierarchical

Historical analyst performance is noisy, sector-dependent, and vulnerable to small-sample overfitting. A simple past-hit-rate multiplier can amplify luck.

For analyst a in sector s :

$$\theta_{a,s} \sim \mathcal{N}(\theta_s, \tau_s^2), \quad (14.1)$$

$$y_{e,h} \sim \mathcal{N}(\theta_{a,s}, \sigma_h^2), \quad (14.2)$$

where $y_{e,h}$ is the realized directionally signed excess return following past events. The point-in-time quality score is:

$$q_{a,s,t} = \mathbb{E}[\theta_{a,s} \mid \mathcal{D}_{<t}]. \quad (14.3)$$

14.2 Separate characteristics

Do not compress the following into one opaque score prematurely:

- sector-specific directional skill;
- timeliness relative to peers;
- boldness or distance from consensus;
- brokerage-firm reputation;
- analyst experience and coverage load;
- forecast stickiness;
- historical sample size and uncertainty.

Stickiness is not automatically bad. A slow-moving analyst may issue fewer but more consequential revisions. Quality enters only after the unweighted analyst signal passes its own out-of-sample gate.

14.3 Shrinkage alternative

A simpler pre-registered approximation is:

$$q_{a,t}^{shrunk} = \bar{y}_{a,t} \frac{N_{a,t}}{N_{a,t} + N_0}, \quad N_0 = 20. \quad (14.4)$$

This remains secondary to the hierarchical model.

Canonical Stock Signal V2

15.1 Institution-deduplicated event sum

The canonical rating-only stock signal uses true upgrades and downgrades, firm-specific normalized scales, one contribution per institution-stock-trading-day, and a 20-day half-life:

$$A_{i,t}^{raw} = \sum_{e \in \mathcal{E}_{i,t}^R} \Delta r_e 2^{-d_e/20}. \quad (15.1)$$

If multiple same-day records exist for one institution, use the final publicly available state or one reconciled economic event. Retain all raw records in lineage.

15.2 Robust sector normalization

Analyst activity and rating distributions vary materially by sector. Define:

$$Z_{i,t}^A = \frac{A_{i,t}^{raw} - \text{median}_{k \in s(i)} A_{k,t}^{raw}}{1.4826 \text{MAD}_{k \in s(i)}(A_{k,t}^{raw}) + \varepsilon}. \quad (15.2)$$

Winsorize the resulting score at pre-registered training-sample percentiles, for example $[-5, 5]$.

15.3 Stock reliability

Let $q_{i,t}^{data} \in [0, 1]$ summarize timestamp quality, rating-map confidence, and entity-mapping quality. Let $N_{i,t}^{eff}$ be the effective contributor count. Define:

$$\rho_{i,t} = \frac{N_{i,t}^{eff}}{N_{i,t}^{eff} + N_0} q_{i,t}^{data}, \quad N_0 = 3. \quad (15.3)$$

The reliable stock signal is:

$$\tilde{A}_{i,t} = Z_{i,t}^A \rho_{i,t}. \quad (15.4)$$

15.4 Independent diagnostics

Store but do not automatically multiply into the canonical score:

$$u_{i,t}, \quad B_{i,t}^A, \quad x_{i,t}^{PT}, \quad x_{i,t}^{EPS}, \quad EventDiversity_{i,t}, \quad q_{a,s,t}.$$

This makes ablation and attribution possible and prevents a large nonlinear formula from hiding weak ingredients.

Stock-First Seed Selection

16.1 Bullish and bearish seeds

At decision time t , use only stocks with valid PIT data, sufficient liquidity, and a mapped ETF membership. Define:

$$\mathcal{S}_t^+ = \left\{ i : \tilde{A}_{i,t} \geq Q_{0.90,t}^{sector} \right\}, \quad (16.1)$$

$$\mathcal{S}_t^- = \left\{ i : \tilde{A}_{i,t} \leq Q_{0.10,t}^{sector} \right\}. \quad (16.2)$$

Alternative fixed thresholds are secondary. Quantile thresholds adapt to changing event intensity and rating coverage.

16.2 Hybrid seed rule

A robust seed rule may require both magnitude and breadth:

$$Seed_i^+ = \mathbb{1} \left[\tilde{A}_{i,t} \geq q_{0.90} \right] \mathbb{1} \left[N_{i,t}^{eff} \geq 2 \right]. \quad (16.3)$$

Test the magnitude-only and magnitude-plus-breadth versions separately. Do not assume the stricter rule is better.

16.3 Neutral control stocks

The research universe must retain neutral stocks and all eligible ETFs. Stock-first discovery is an efficient candidate-generation mechanism, not permission to redefine the comparison universe after observing the signal.

Point-in-Time Stock-to-ETF Discovery

17.1 Reverse index

Build a point-in-time reverse map:

$$(t, i) \longrightarrow \{(j, w_{ij,t}, available_{ij,t})\}.$$

For each decision date, candidate ETFs are:

$$\mathcal{E}_t^{cand} = \left[\bigcup_{i \in \mathcal{S}_t^+ \cup \mathcal{S}_t^-} \{j : w_{ij,t} > 0\} \right] \cap EligibleETFs_t. \quad (17.1)$$

Only holdings snapshots publicly available by t may enter the reverse index. QuantConnect's documented possible one-week delay requires an explicit sensitivity test [2].

17.2 ETF eligibility

Canonical eligibility at each date:

- US-listed long equity ETF;
- ETF age at least one year;
- trailing median daily dollar volume at least \$5 million;
- point-in-time AUM at least \$100 million when reliable historical AUM exists;
- no leveraged, inverse, volatility, commodity, bond, crypto, or closed-end products in alpha discovery;
- at least 99% candidate ETF weight mapped to valid securities;
- holdings snapshot passes completeness tests.

If historical AUM is not trustworthy, use liquidity, age, and price filters as the canonical gate and move AUM to a robustness test.

17.3 Holdings availability sensitivity

Run at least:

$$H_0 : available = LastUpdate, \quad (17.2)$$

$$H_1 : available = \max(EndTime, LastUpdate) + 1 \text{ trading day}, \quad (17.3)$$

$$H_5 : available = EndTime + 5 \text{ trading days}. \quad (17.4)$$

A signal that disappears under a conservative holdings lag is not robust enough for promotion.

ETF Aggregation, Coverage, and Reliability

18.1 Covered weight

Let $O_{i,t} = 1$ if a valid stock signal exists. Define:

$$C_{j,t} = \sum_i w_{ij,t} O_{i,t}. \quad (18.1)$$

18.2 Coverage-normalized magnitude

$$\bar{S}_{j,t} = \frac{\sum_i w_{ij,t} \tilde{A}_{i,t}}{\max(C_{j,t}, \varepsilon)}. \quad (18.2)$$

This asks: among the covered portion of the ETF, how strong is analyst evidence?

18.3 Contribution concentration

Let:

$$p_{ij,t} = \frac{|w_{ij,t} \tilde{A}_{i,t}|}{\sum_k |w_{kj,t} \tilde{A}_{k,t}| + \varepsilon}. \quad (18.3)$$

The effective number of contributing stocks is:

$$N_{j,t}^{eff} = \frac{1}{\sum_i p_{ij,t}^2}. \quad (18.4)$$

18.4 ETF reliability shrinkage

$$Rel_{j,t} = \sqrt{C_{j,t}} \min \left(1, \sqrt{\frac{N_{j,t}^{eff}}{N_0^{ETF}}} \right), \quad N_0^{ETF} = 5. \quad (18.5)$$

The canonical ETF score is:

$$S_{j,t}^{V2} = \bar{S}_{j,t} Rel_{j,t}. \quad (18.6)$$

Coverage thresholds 30%, 50%, and 70% remain robustness tests. The continuous reliability function is the preferred main specification.

ETF Breadth, Concentration, and Peer Residualization

19.1 Directional breadth

$$Breadth_{j,t}^{ETF} = \frac{\sum_i w_{ij,t} \text{sign}(\tilde{A}_{i,t}) O_{i,t}}{\max(C_{j,t}, \varepsilon)}. \quad (19.1)$$

Also report bullish and bearish covered-weight breadth separately:

$$B_{j,t}^+ = \frac{\sum_i w_{ij,t} \mathbb{1}[\tilde{A}_{i,t} > 0]}{\max(C_{j,t}, \varepsilon)}, \quad (19.2)$$

$$B_{j,t}^- = \frac{\sum_i w_{ij,t} \mathbb{1}[\tilde{A}_{i,t} < 0]}{\max(C_{j,t}, \varepsilon)}. \quad (19.3)$$

Do not automatically multiply breadth by magnitude in V2. Run two-dimensional sorts.

19.2 ETF holdings concentration

Measure conventional holdings concentration:

$$HHI_{j,t}^{holdings} = \sum_i w_{ij,t}^2. \quad (19.4)$$

Measure signal-contribution concentration:

$$HHI_{j,t}^{signal} = \sum_i p_{ij,t}^2. \quad (19.5)$$

An ETF may be diversified by holdings but still have an analyst signal dominated by one constituent.

19.3 Peer-relative score

Create ETF peer groups from holdings similarity, category, and dominant exposure. Then:

$$Z_{j,t}^{peer} = \frac{S_{j,t}^{V2} - \text{median}_{k \in peer(j)} S_{k,t}^{V2}}{1.4826 \text{MAD}_{k \in peer(j)}(S_{k,t}^{V2}) + \varepsilon}. \quad (19.6)$$

19.4 Residual score

An advanced diagnostic residualizes known ETF exposures:

$$S_{j,t}^{\perp} = S_{j,t}^{V2} - \hat{\beta}_t^{\top} \mathbf{X}_{j,t}, \quad (19.7)$$

where $\mathbf{X}$ includes sector exposures, 20-day and 12-1 momentum, beta, volatility, value/-growth, market-cap tilt, liquidity, and concentration. Coefficients must be estimated using training data only.

Execution, Rebalancing, and Portfolio Rules

20.1 Event-driven clock

Update analyst stock signals and ETF scores every trading day after all available events are ingested. Execute at the next eligible market open.

20.2 Hysteresis

To limit turnover, use separate entry and exit thresholds:

$$EnterLong_{j,t} = \mathbb{I}[Percentile(Z_{j,t}^{peer}) \geq 90], \quad (20.1)$$

$$ExitLong_{j,t} = \mathbb{I}[Percentile(Z_{j,t}^{peer}) < 70]. \quad (20.2)$$

For defensive use:

$$Underweight_{j,t} = \mathbb{I}[Percentile(Z_{j,t}^{peer}) \leq 10], \quad (20.3)$$

$$Restore_{j,t} = \mathbb{I}[Percentile(Z_{j,t}^{peer}) > 30]. \quad (20.4)$$

Test daily event-driven, weekly, and monthly schedules. Daily with hysteresis is primary; weekly is the primary turnover-control robustness test.

20.3 Initial portfolio construction

The first investable long-only portfolio:

- select one representative ETF per high-overlap cluster;
- hold the top five eligible ETFs by $Z_{j,t}^{peer}$;
- equal weight or inverse-volatility weight;
- maximum ETF weight 20%;

- maximum sector exposure 40%;
- maximum holdings-cluster exposure 30%;
- no leveraged or inverse products.

A score-weighted version is secondary:

$$\omega_{j,t}^{raw} = \frac{\max(Z_{j,t}^{peer}, 0)}{\sum_k \max(Z_{k,t}^{peer}, 0)}. \quad (20.5)$$

20.4 Volatility target

$$\kappa_t = \min\left(1, \frac{\sigma^{target}}{\hat{\sigma}_{p,t}}\right), \quad \omega_t = \kappa_t \omega_t^{raw}. \quad (20.6)$$

Cash is the default residual asset. Leverage is a later gated experiment.

Backtest Program and Experiment Rounds

21.1 Research sequence

Round	Purpose	Promotion requirement
0	Data feasibility and PIT audit	Valid timestamps, mappings, holdings lags, and sufficient coverage
1	Stock-level event study	At least one analyst event channel predicts realistic executable returns after controls
2	Stock-signal battery	Freshness, sector normalization, breadth, and novelty show stable OOS behavior
3	Aggregation topology	ETF aggregation adds value versus direct stocks and industry aggregation
4	ETF cross-sectional validation	Positive OOS IC and economically meaningful monotonic spread
5	Investable long-only portfolio	Net performance survives costs, overlap controls, and turnover limits
6	Analyst-quality extensions	Quality features add incremental OOS value without concentration
7	Three-engine integration	Analyst engine adds marginal value to insider and short-interest ensemble
8	Leverage research	Only after unleveraged analyst sleeve passes all gates

21.2 Walk-forward design

For a sample beginning in 2015, a canonical rolling design is:

```

Train:    5 years
Validate: 2 years
Test:     1 year
Step:     1 year

```

If the sample is too short, reduce model complexity before reducing the holdout discipline.

Never call 2024–2026 an untouched holdout if those years have already influenced feature selection.

21.3 Primary horizons

Stock event study:

1d, 5d, 20d, 60d.

ETF ranking:

5d, 20d, 60d.

The 20-day horizon is primary because it matches the canonical rating half-life and allows a practical ETF allocation cycle.

Statistical Inference and Confounder Controls

22.1 Information coefficient

At each decision date:

$$IC_t^{(h)} = \text{SpearmanCorr} \left(Z_{j,t}^{peer}, R_{j,t \rightarrow t+h}^{excess} \right). \quad (22.1)$$

Report mean IC, median IC, ICIR, positive-period share, and HAC t-statistics.

22.2 Fama-MacBeth-style regression

$$R_{j,t \rightarrow t+h}^{excess} = \alpha_t + \beta_{A,t} Z_{j,t}^{peer} + \gamma_t^\top \mathbf{X}_{j,t} + \mu_{peer(j),t} + \varepsilon_{j,t+h}. \quad (22.2)$$

Average coefficients through time and use HAC errors with lag related to horizon h .

22.3 Mandatory controls

$\mathbf{X}_{j,t}$ includes:

- ETF 20-day, 60-day, and 12-1 momentum;
- sector and industry momentum;
- earnings-event and guidance proximity;
- immediate event price jump;
- market beta and realized volatility;
- value/growth and size tilt;
- liquidity, turnover, and holdings concentration;
- analyst coverage and event intensity.

22.4 Common-event samples

Run:

- all events;
- exact earnings-day reactions;
- 1–2 days after earnings;
- 3–5 days after earnings;
- more than 5 days from earnings;
- pre-earnings revisions.

Simple ± 2 and ± 5 exclusions remain summary robustness checks.

22.5 Multiple testing

Classify every specification as PRIMARY, SECONDARY, or EXPLORATORY. Apply false-discovery-rate control to large batteries and report a deflated Sharpe or comparable data-snooping adjustment for the final selected strategy.

Direct Stock vs Industry vs ETF Tests

This chapter determines whether the ETF layer is useful rather than merely aesthetically appealing.

23.1 Direct seed-stock portfolio

$$P_t^{Stock} = \sum_{i \in S_t^+} \omega_{i,t}^{Stock} R_{i,t}. \quad (23.1)$$

Test equal weight, inverse volatility, and score weight.

23.2 Industry aggregation

For industry g :

$$S_{g,t}^{Industry} = \sum_{i \in g} \omega_{ig,t} \tilde{A}_{i,t}. \quad (23.2)$$

Industry-level aggregate recommendation changes have prior research support and are the correct benchmark for the ETF innovation [5, 6].

23.3 ETF aggregation

$$S_{j,t}^{ETF} = S_{j,t}^{V2}.$$

23.4 Decision metrics

Compare:

- alpha and IC;
- Sharpe and Sortino;

- maximum drawdown and expected shortfall;
- turnover and transaction costs;
- single-stock concentration;
- sensitivity to one event or one company;
- capacity and implementation simplicity.

ETF layer passes only if

ETF aggregation produces greater predictive power, materially better risk-adjusted performance, lower idiosyncratic risk, lower turnover, or superior implementability than direct stocks and conventional industry aggregation. Otherwise, the analyst engine should operate at the stock or industry layer.

Costs, Liquidity, Turnover, and Realism

24.1 Cost model

For portfolio turnover TO_t , spread estimate s_t , commission c_t , and market-impact coefficient ϕ :

$$Cost_t = TO_t(c_t + s_t) + \phi \sum_j \left(\frac{|\Delta DollarPosition_{j,t}|}{ADV_{j,t}} \right)^{1/2}. \quad (24.1)$$

Run at least 0, 5, 10, and 20 bps per side, plus liquidity-sensitive estimates when available. The primary result uses 10 bps per side unless observed spreads support a different pre-registered value.

24.2 Turnover diagnostics

Report:

- daily, monthly, and annual turnover;
- fraction of trades caused by new information versus threshold churn;
- average holding period;
- return captured before and after entry latency;
- turnover by analyst event type;
- turnover reduction from hysteresis and weekly rebalancing.

24.3 Public-feed realism

The main strategy assumes access to the public or vendor-distributed timestamp, not privileged brokerage-client delivery. If earlier subscriber timestamps are unavailable, no back-test may assume them.

Multi-Engine Integration

The analyst engine is one evidence channel in the broader system.

25.1 Standard output

For ETF j , emit:

$$\left(Z_{j,t}^{Analyst}, Confidence_{j,t}^{Analyst}, Breadth_{j,t}^{Analyst}, Freshness_{j,t}, EventComposition_{j,t} \right).$$

Confidence may be:

$$Confidence_{j,t}^A = Rel_{j,t} \times q_{j,t}^{data} \times \min\left(1, \frac{EventDiversity_{j,t}}{D_0}\right). \quad (25.1)$$

25.2 Incremental-value regression

With insider and short-interest scores:

$$R_{j,t \rightarrow t+h}^{excess} = \alpha_t + \beta_A Z_{j,t}^A + \beta_I Z_{j,t}^I + \beta_S Z_{j,t}^S + \gamma^\top \mathbf{X}_{j,t} + \varepsilon_{j,t+h}. \quad (25.2)$$

The analyst sleeve receives capital only if β_A or its portfolio contribution remains economically meaningful out of sample.

25.3 Late fusion is canonical

Each engine produces an independently validated ETF portfolio. Combine after attribution:

$$\mathbf{w}_t^{combined} = \alpha_A \mathbf{w}_t^A + \alpha_I \mathbf{w}_t^I + \alpha_S \mathbf{w}_t^S, \quad (25.3)$$

subject to strategy-risk budgets and portfolio exposure constraints. Early fusion over raw data is exploratory only.

Implementation Blueprint and Pseudocode

26.1 Suggested repository structure

```
alphas/  
  analyst_etf/  
    ontology.py  
    event_taxonomy.py  
    consensus.py  
    novelty.py  
    target_revisions.py  
    eps_revisions.py  
    freshness.py  
    breadth.py  
    quality.py  
    stock_signal.py  
    reverse_index.py  
    etf_aggregation.py  
    peer_groups.py  
    portfolio.py  
    diagnostics.py  
  
data/  
  raw/analyst_events/  
  normalized/analyst_events/  
  normalized/security_master/  
  normalized/etf_holdings/  
  derived/stock_signals/  
  derived/etf_signals/  
research/analyst_etf_v2/  
  preregistration/  
  configs/  
  experiments/  
  reports/  
tests/  
  test_rating_ontology.py  
  test_no_lookahead.py  
  test_consensus_excludes_event_analyst.py  
  test_etf_holdings_pit.py  
  test_security_mapping.py  
  test_event_deduplication.py  
  test_holdings_lag.py
```

26.2 Pseudocode

```

def build_stock_signal(as_of, events, ontology, security_master):
    valid = events.available_by(as_of)
    valid = map_entities(valid, security_master, as_of)
    valid = classify_event_types(valid)
    valid = normalize_with_firm_scales(valid, ontology, as_of)
    valid = dedupe_institution_stock_day(valid)

    rating_events = valid.true_rating_revisions()
    rating_events["age"] = trading_days(rating_events.time, as_of)
    rating_events["contribution"] = (
        rating_events.delta_rating
        * 2 ** (-rating_events.age / 20.0)
    )

    raw = rating_events.groupby("security_id").contribution.sum()
    sector_z = robust_sector_zscore(raw, as_of)
    n_eff = effective_contributor_count(rating_events)
    data_quality = compute_data_quality(rating_events)
    reliability = n_eff / (n_eff + 3.0) * data_quality
    return sector_z * reliability

def discover_and_score_etfs(as_of, stock_signal, reverse_index, eligibility):
    seeds = select_quantile_seeds(stock_signal, as_of, 0.10, 0.90)
    candidates = union(reverse_index.lookup(as_of, s) for s in seeds.index)
    candidates = candidates.intersection(eligibility(as_of))
    rows = join_pit_holdings(candidates, stock_signal, as_of)
    return aggregate_with_reliability(rows, min_effective_contributors=5)

```

26.3 QuantConnect integration

Use QuantConnect for historical ETF universes, price data, symbol mapping, corporate actions, and order simulation. Import analyst events as immutable custom data or a pre-computed daily stock-signal table. The algorithm consumes only normalized, versioned records.

Acceptance Gates, Risk Register, and Final Recommendation

27.1 Data gate

Status	Condition
GO	Exact or conservative analyst availability; firm ontology audited; historical ETF holdings and lag enforced; 99% candidate weight mapped; survivorship handling documented
REMEDiate	Adequate coverage but one noncanonical field family, such as EPS revisions, is weak; continue with simpler rating-only V2
NO-GO	Current tickers, future holdings, unknown publication clocks, unreconciled duplicate events, or retrospective consensus reconstruction contaminate the sample

27.2 Signal gate

Promote to ETF portfolio construction only if:

- stock-level executable event returns are positive at a primary horizon after controls;
- firm-specific normalization is at least as strong as the global map;
- OOS ETF IC is positive and reasonably stable;
- top-minus-bottom spreads are economically meaningful and approximately monotonic;
- the result survives momentum, earnings, peer, overlap, and holdings-lag controls;
- ETF aggregation adds value relative to direct stocks and industry aggregation;
- realistic costs do not eliminate the edge.

27.3 Ensemble gate

The analyst engine receives a persistent risk budget only if it adds marginal OOS value to the insider and short-interest ensemble. If it merely repeats momentum or earnings information, retain it as a diagnostic confirmation feature rather than as an independent capital sleeve.

27.4 Risk register

Risk	Severity	Mitigation
Publication-time lookahead	Critical	Availability timestamp and next-open execution; conservative date-only cohort
Firm-scale misclassification	Critical	Date-aware ontology, manual audit, quarantined ambiguous labels
Historical consensus left censoring	High	Warm-up period, active-rating expiry, benchmark-only use
Common earnings-event double counting	High	Event IDs, conditional samples, event-diversity diagnostics
Ticker/security mismatch	Critical	Permanent security master, historical validity intervals
ETF holdings lag	Critical	LastUpdate/EndTime plus 1- and 5-day lag tests
Momentum repackaging	High	Stock and ETF momentum controls; residual score; direct comparison
Overlapping ETF duplicates	High	Holdings similarity clusters and one representative ETF per cluster
Analyst-quality overfit	High	Hierarchical shrinkage, late-stage-only inclusion, nested walk-forward
Sparse bearish ratings	Medium	Analyze bullish and bearish sides separately; do not force symmetry
Vendor history revisions	High	Immutable raw snapshots, checksums, and data-version hashes
Multiple testing	High	Pre-registration, FDR, untouched holdout, deflated Sharpe

Final recommendation

Implement the rating-only canonical V2 first: firm-normalized revisions, 20-day decay, institution-day deduplication, robust sector normalization, stock reliability, stock-first ETF discovery, coverage-normalized ETF magnitude, and effective-contributor reliability. Keep novelty, price-target, EPS, breadth, and analyst quality as separate experiments. The decisive research round is the direct-stock versus industry versus ETF comparison.

Appendix A: Data Dictionary

Field	Type	Definition
event_id	string	Vendor or canonical event identifier
security_id	string	Permanent internal security key
issuer_id	string	Permanent issuer key
analyst_id	string	Canonical analyst identity, nullable
institution_id	string	Canonical brokerage-firm identity
event_timestamp	timestamp	Timestamp associated with analyst action
available_timestamp	timestamp	Earliest timestamp permitted for strategy use
action_type	category	Upgrade, downgrade, initiation, resumption, reiteration, maintain, target-only, termination
rating_prior_raw	string	Vendor-provided prior language
rating_current_raw	string	Vendor-provided current language
rating_prior_norm	float	Firm-specific normalized prior rating
rating_current_norm	float	Firm-specific normalized current rating
delta_rating	float	Current normalized rating minus prior normalized rating
consensus_excl_analyst	float	Freshness-weighted pre-event consensus excluding contributor
novelty	float	New rating minus excluded-contributor consensus
pt_prior, pt_current	float	Prior/current price targets
pt_revision_price_scaled	float	Target revision divided by pre-event price
eps_prior, eps_current	float	Point-in-time analyst EPS estimates for a specified horizon
common_event_id	string	Earnings, guidance, filing, price-gap, or industry-event cluster
mapping_quality	float	Confidence in entity and rating mapping
timestamp_quality	float	Exact, session-level, date-only, or invalid

raw_payload_hash	string	Immutable vendor-payload checksum
data_snapshot_id	string	Version of raw/normalized data used

Appendix B: Experiment IDs and Configuration

29.1 Experiment names

```
ANETF2-R0-DATA-AUDIT
ANETF2-R1-GLOBAL-MAP
ANETF2-R1-FIRM-MAP
ANETF2-R1-NOVELTY
ANETF2-R1-PRICE-TARGET
ANETF2-R1-EPS-REVISION
ANETF2-R2-DECAY05
ANETF2-R2-DECAY20
ANETF2-R2-DECAY60
ANETF2-R2-SECTOR-NORM
ANETF2-R2-BREADTH
ANETF2-R3-DIRECT-STOCK
ANETF2-R3-INDUSTRY
ANETF2-R3-ETF-RAW
ANETF2-R3-ETF-RELIABILITY
ANETF2-R3-ETF-PEER
ANETF2-R4-CROSS-SECTION
ANETF2-R5-PORTFOLIO
ANETF2-R6-ANALYST-QUALITY
ANETF2-R7-ENSEMBLE
ANETF2-R8-LEVERAGE
```

29.2 Canonical YAML

```
experiment_id: ANETF2-R3-ETF-RELIABILITY
status: PRIMARY
sample:
  start: 2015-01-01
  end: null
  warmup_months: 12
analyst_signal:
  firm_specific_rating_scale: true
  event_types: [upgrade, downgrade]
  institution_stock_day_dedupe: true
  half_life_trading_days: 20
```

```
  sector_robust_zscore: true
  stock_reliability_n0: 3
etf_aggregation:
  stock_first_discovery: true
  coverage_normalized: true
  etf_effective_contributor_target: 5
  holdings_lag_trading_days: 1
  peer_relative_rank: true
execution:
  rule: next_open
  date_only_extra_delay_days: 1
  rebalance: event_driven_daily
  entry_percentile: 90
  exit_percentile: 70
costs:
  bps_per_side: 10
portfolio:
  max_etf_weight: 0.20
  max_sector_weight: 0.40
  max_overlap_cluster_weight: 0.30
  leverage: false
validation:
  train_years: 5
  validate_years: 2
  test_years: 1
  multiple_testing: FDR
```

Appendix C: Required Outputs

30.1 Standard tables

Every major experiment produces:

1. stock event returns by event type, timing cohort, and horizon;
2. ETF quintile and decile returns at 5, 20, and 60 days;
3. Q1–Q5 spread with HAC and block-bootstrap confidence intervals;
4. IC, ICIR, positive-period frequency, and year-by-year IC;
5. direct-stock versus industry versus ETF comparison;
6. global versus peer-relative ranking comparison;
7. breadth-by-magnitude two-dimensional sort;
8. coverage and effective-contributor buckets;
9. portfolio performance before and after costs;
10. regime, sector, ETF-category, and overlap-cluster attribution;
11. incremental-value regression with insider and short-interest signals.

30.2 Required plots

1. event-time cumulative abnormal return by rating action;
2. signal percentile versus future excess return;
3. ETF quintile/decile monotonicity;
4. cumulative Q1–Q5 spread;
5. rolling IC and rolling Sharpe;
6. year-by-year OOS alpha;
7. signal decay by horizon;
8. breadth versus magnitude heatmap;

9. effective contributors versus return;
10. coverage versus reliability-adjusted performance;
11. stock, industry, and ETF cumulative comparison;
12. turnover versus net return;
13. drawdown and time underwater;
14. contribution by ETF holdings cluster;
15. analyst-signal marginal contribution in the three-engine ensemble.

30.3 Disposition template

Each report ends with exactly one disposition:

<i>PASS</i> <i>FAIL</i> <i>INCONCLUSIVE</i> <i>INVALID-DATA</i>

and one permitted next action:

- advance to the next pre-registered round;
- repeat only after a documented data correction;
- create a separately labeled exploratory hypothesis;
- stop the research branch.

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